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Site-Specific Probing for Full-Duplex Millimeter-Wave Systems

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Self-interference in full-duplex mmWave systems

- Recent efforts dedicated to equipping millimeter-wave (mmWave) communication systems with full-duplex capabilities have been spearheaded by beamforming techniques that can spatially mitigate the consequent self-interference (SI) coupling that manifests.
- Works such as STEER [1] and STEER+ [2] have sought to address the challenge of mitigating self-interference by leveraging a joint selection of transmit and receive beams designed to ensure high gain in the desired steering directions while maintaining low self-interference coupling.
- LONESTAR codebooks were designed in [3] to minimize self-interference coupling for every transmit-receive beam pair within the codebooks. These codebooks rely on good estimates of the SI channel.
- Ultimately, the task of efficiently and reliably estimating self-interference is a difficult one and remains unsolved, to the best of our knowledge.
- Estimating SI while accurately conducting beam alignment via traditional methods poses further difficulties.

[1] I. P. Roberts, A. Chopra, T. Novlan, S. Vishwanath, and J. G. Andrews, "STEER: Beam selection for full-duplex millimeter wave communication systems," *IEEE Trans. Commun.*, vol. 70, no. 10, pp. 6902–6917, 2022.
[2] I. P. Roberts, Y. Zhang, T. Osman, and A. Alkhateeb, "STEER+: Robust beam refinement for full-duplex millimeter wave communication systems," in *Proc. Asilomar Conf. Signals, Sys., and Comput.*, Oct. 2023.
[3] I. P. Roberts, S. Vishwanath, and J. G. Andrews, "LoneSTAR: Analog beamforming codebooks for full-duplex millimeter wave systems," *IEEE Transactions on Wireless Communications*, vol. 22, no. 9, pp. 5754–5769, Sep. 2023.

A joint beam alignment and channel estimation scheme

- Inspired by [1] and [2], we equip the BS with a trainable probing codebook $\mathcal{F}$ to learn the site-specific characteristics of the environment.
- The probing and sensing codebooks $\mathcal{F}$ and $\mathcal{W}$ are used to collect measurements of self-interference channel.
- The proposed method:
  1) The BS sweeps over the probing codebook.
  2) The user returns power measurements back to the BS.
  3) Codebooks $\mathcal{F}$ and $\mathcal{W}$ are used to obtain a composite received signal.
  4) The power measurements are used to optimize beam alignment via a beam selection function f.
  5) The composite received signal is used by the self-interference channel estimator g to estimate SI coupling.

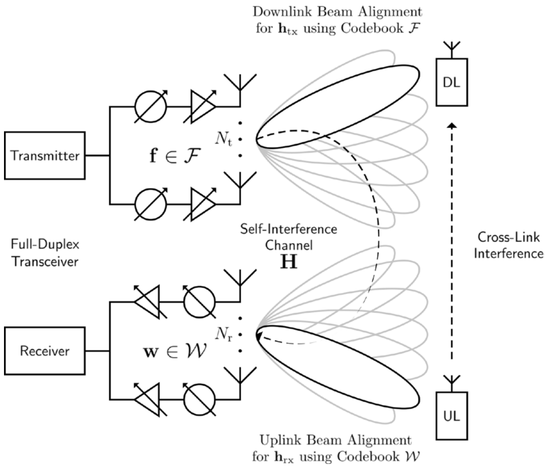
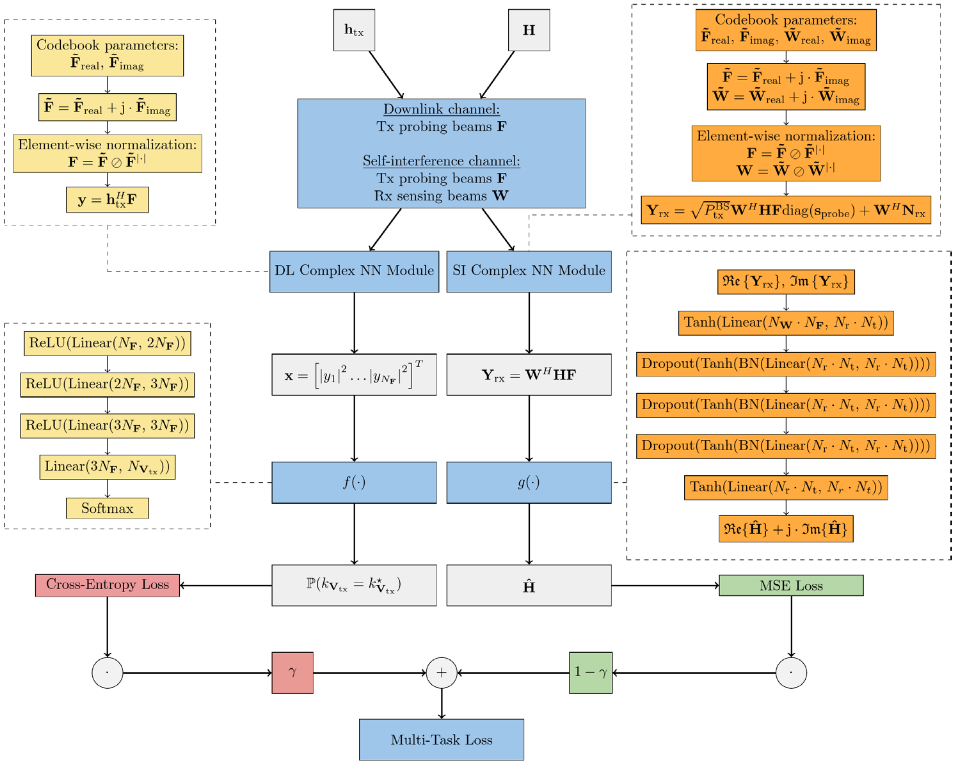


Illustration of the DeepMIMO O1 Scenario. Adapted from [1].

[1] Y. Heng, J. Mo, and J. G. Andrews, “Learning site-specific probing beams for fast mmWave beam alignment,” *IEEE Transactions on Wireless Communications*, vol. 21, no. 8, pp. 5785–5800, Aug. 2022.
[2] Y. Heng and J. G. Andrews, “Grid-free MIMO beam alignment through site-specific deep learning,” *IEEE Transactions on Wireless Communications*, vol. 23, no. 2, pp. 908–921, Feb. 2024.

A multi-task neural network architecture

- Normalized realizations of the BS-user channels and realizations of the SI channel are used to train the NN.
- Cross-entropy loss is used to optimize the beam selection function f.
- Mean squared error (MSE) is used to optimize the self-interference channel estimator g.
- The mappings f and g are simple multilayer perceptrons (MLPs).
- The hyperparameter γ throttles the trade-off between optimal narrow beam selection and the estimation of self-interference coupling.



Experimental setup

- The BS-user channels follow a ray-based model and are generated using [1]. The Rician SI channels are generated using [2].
- We consider the line-of-sight (LOS) DeepMIMO O1 Scenario.
- Two-stage hierarchical search baseline: $\mathcal{F}$ is now a discrete Fourier transform (DFT) codebook.

DEEPMIMO SIMULATION PARAMETERS FOR THE BS-UE CHANNELS

Tx antenna	64 × 1 ULA
UE antenna	Single
Narrow beam codebook size	128
Carrier frequency	28 GHz
Bandwidth	100 MHz
Number of rays (N_{p})	12

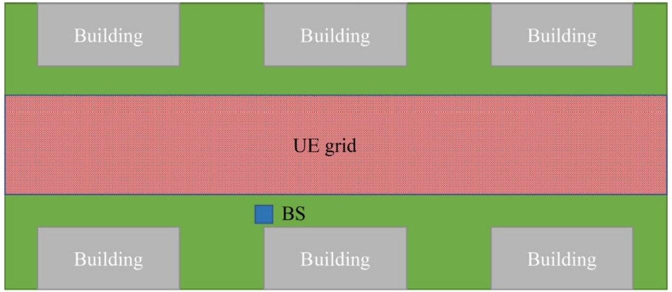


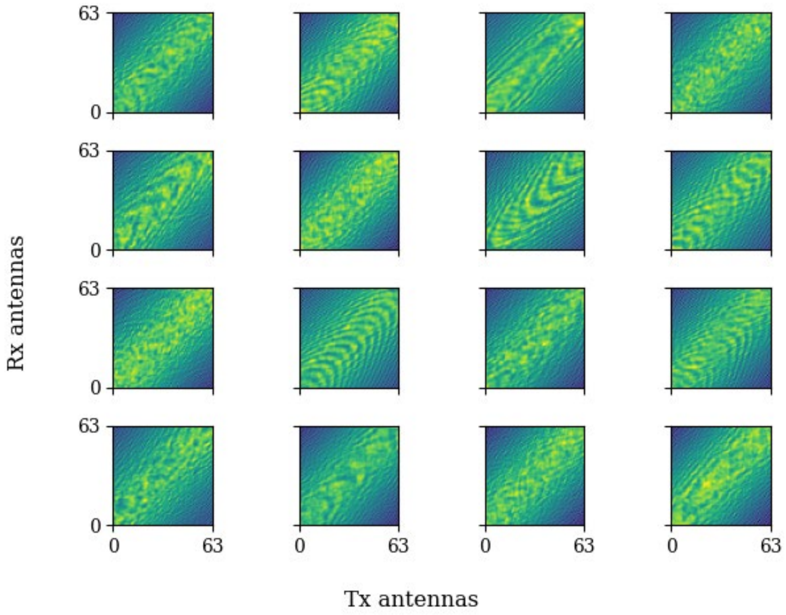
Illustration of the DeepMIMO O1 Scenario [3].

SIMULATION PARAMETERS FOR SELF-INTERFERENCE COUPLING

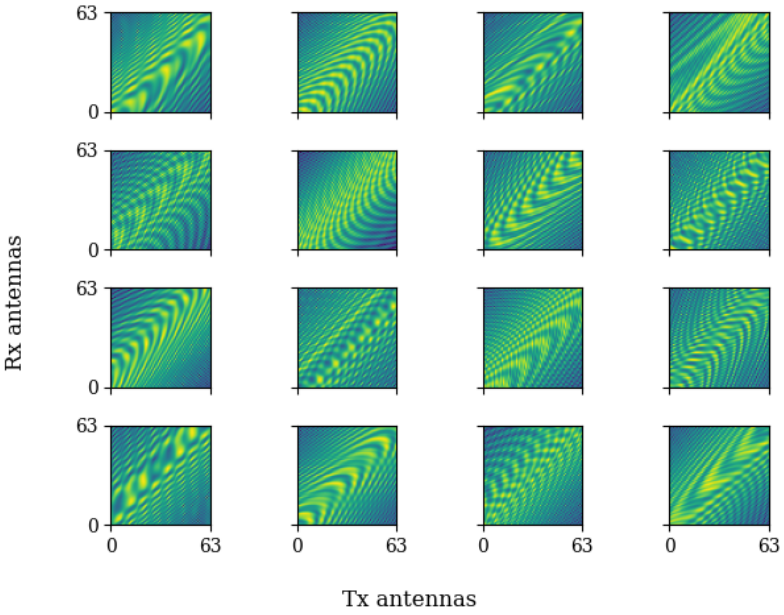
Tx antenna	64 × 1 ULA
Rx antenna	64 × 1 ULA
Carrier frequency	28 GHz
Bandwidth	100 MHz
Number of rays (M_{p})	3
Rice factor (κ)	10 dB
Spacing between Tx and Rx arrays	20λ
Azimuth spread (AoA/AoD)	$[-\frac{\pi}{4}, \frac{\pi}{4}]$
Elevation spread (AoA/AoD)	$[-\frac{\pi}{2}, \frac{\pi}{2}]$

[1] A. Alkhateeb, “DeepMIMO: A generic deep learning dataset for millimeter wave and massive MIMO applications,” in *Proc. of Information Theory and Applications Workshop (ITA)*, San Diego, CA, Feb. 2019, pp. 1–8.
[2] I. P. Roberts, “MIMO for MATLAB: A toolbox for simulating MIMO communication systems,” Nov. 2021. [Online]. Available: https://arxiv.org/abs/2111.05273
[3] Y. Heng and J. G. Andrews, “Grid-free MIMO beam alignment through site-specific deep learning,” *IEEE Transactions on Wireless Communications*, vol. 23, no. 2, pp. 908–921, Feb. 2024.

Sample self-interference channel estimates



Selected estimates of the SI channel generated by our algorithm (γ = 0.6).



Actual realizations of the self-interference channel.

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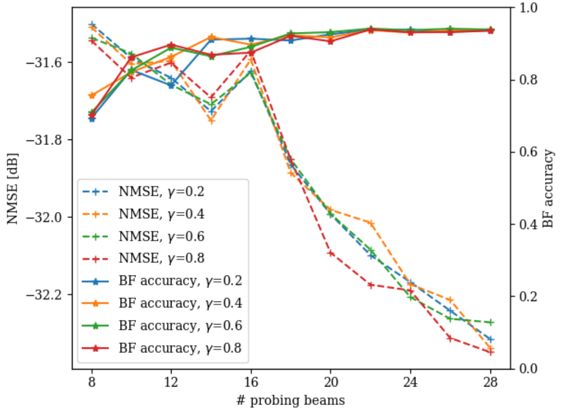
Accuracy of beam alignment selection and self-interference channel estimates

- Normalized mean squared error (NMSE) is used to evaluate the accuracy of the SI estimates.
- Utilizing $\mathcal{F}$ as site-specific probing codebook rather than a DFT codebook yields better beam alignment accuracies at no cost to the accuracy of the estimates of self-interference channel.
- Increasing the number of probing beams $|\mathcal{F}|$ (as well as the number of sensing beams) improves estimation of the SI channel.
- Our algorithm yields low NMSE scores and still maintains high beam alignment accuracies when $|\mathcal{F}|$ is sufficiently large.
- Our results might suggest that when the BS uses enough probing (pilot) resources during training and can no longer refine its optimal beam selection capabilities, the BS can use its remaining probing resources to provide better estimates for the self-interference coupling.



SI NMSE and beam alignment accuracy vs. $|\mathcal{F}|$ for $\gamma \in \{0.2, 0.4, 0.6, 0.8\}$.

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Conclusion

- We develop a novel algorithm that simultaneously performs beam alignment and self-interference channel estimation algorithm in a full-duplex mmWave setting.
- The trainable probing codebook can capture important characteristics about the propagation environment. It can also be simultaneously harnessed (together with a sensing codebook) by a neural network to generate accurate estimates of the self-interference channel.
- Future works could assess whether our proposed joint beam alignment and channel estimation algorithm still performs well in environments characterized by many non-line-of-sight (NLOS) users.
- Ultimately, our study motivates a variety of future works related to self-interference estimation and cancellation during the beam alignment process.